

Zhuohang Jiang

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Profile

I am a **Ph.D. candidate in Computer Science** at **The Hong Kong Polytechnic University**, advised by **Prof. Qing Li** and **Prof. Wenqi Fan**. My research focuses on **Large Language Models**, **Retrieval-Augmented Generation**, **multimodal agents**, and **recommender systems**, with an emphasis on building reliable LLM-based systems for hierarchical reasoning, multimodal knowledge-intensive question answering, and industrial cross-domain recommendation. My current cumulative GPA is **3.60/4.00**.

Previously, I received my **B.Eng. in Computer Science and Technology** from **Sichuan University** in 2024, with a major GPA of **3.79/4.00** and an overall GPA of **3.78/4.00**. I worked as a research assistant at **MachineLab / DICALab**, advised by **Prof. JiZhe Zhou**, where I participated in projects supported by the **National Natural Science Foundation of China** and the **National Key R&D Program of China**.

Research Summary

My recent work studies how to connect the semantic reasoning ability of large foundation models with practical systems. Representative directions include **LLM reasoning benchmarks** such as **HiBench**, **multimodal RAG and smart-glasses agents** such as **QA-Dragon** and **SUPER-GLASSES/SUPERLENS**, and **LLM-enhanced recommender systems** such as **AIR** for industrial cross-domain recommendation. My earlier research focused on computer vision, especially **image manipulation detection and localization** and **privacy-sensitive object recognition**. My publications include **KDD 2026**, **CVPR 2026 Findings**, **KDD 2025**, **NeurIPS 2024 Spotlight**, **AAAI 2025**, and **ICONIP 2023**; my work has received **500+ citations** with an **h-index of 7**.

Education

The Hong Kong Polytechnic University

PH.D. IN COMPUTER SCIENCE

Hong Kong SAR

Sep. 2024 – Present

- Advisors: Prof. Qing Li and Prof. Wenqi Fan.
- Cumulative GPA: 3.60/4.00.
- Selected coursework: Advanced Machine Learning (A), Advanced Topics in Optimization (A-), Advanced Topics in Visual Computing (A-).
- Completed: Research Seminar I & II, Practicum I, Engineering Ethics, Presentation Skills.

Sichuan University

B.ENG. IN COMPUTER SCIENCE AND TECHNOLOGY

Chengdu, China

Sep. 2020 – Jun. 2024

- Major GPA: 3.79/4.00 (89.39/100); Overall GPA: 3.78/4.00 (89.25/100).
- Research Assistant at MachineLab / DICALab, Sichuan University, 2022–2024.

Selected Publications

- *A Survey on AI Smart Glasses for Wearable Intelligence: From Egocentric Sensing to Agentic Personalization*
X. Yuan, Y. Wang, **Z. Jiang**, H. Qu, Y. Ding, S. Lin, G. Xing, H. Yang, J. Cao, Q. Li, W. Fan. **Preprints.org preprint**, 202608.0648, 2026.
- *A/B Agent: A Self-Evolving Agent for Strategy Iteration in Industrial A/B Testing*
Z. Jiang, Y. Chen, Y. Pan, Z. Hu, W. Fan, Q. Li, H. Wang, J. Wang, W. Ou. **arXiv preprint**, arXiv:2608.04625, 2026.
- *Atomic Intent Reasoning: Bringing LLM Semantics to Industrial Cross-Domain Recommendations*
Z. Jiang, Y. Chen, S. Wang, H. Qu, J. Zhou, W. Fan, Q. Li, D. Liang, J. Wang. **KDD 2026**, Applied Data Science Track. CCF-A.
- *SUPERGLASSES: Benchmarking Vision Language Models as Intelligent Agents for AI Smart Glasses*
Z. Jiang, X. Yuan, H. Qu, S. Lin, K. Liu, W. Fan, Q. Li. **CVPR 2026 Findings**. CCF-A.
- *QA-Dragon: Query-Aware Dynamic RAG System for Knowledge-Intensive Visual Question Answering*
Z. Jiang, P. Wu, X. Yuan, W. Fan, Q. Li. **KDD 2025 Workshop on Multimodal Retrieval Augmented Generation**. CCF-A.
- *A Survey of WebAgents: Towards Next-Generation AI Agents for Web Automation with Large Foundation Models*
L. Ning, Z. Liang, **Z. Jiang**, H. Qu, Y. Ding, W. Fan, X.-Y. Wei, S. Lin, H. Liu, P. S. Yu, Q. Li. **KDD 2025 Tutorial Track**. CCF-A.
- *HiBench: Benchmarking LLMs Capability on Hierarchical Structure Reasoning*
Z. Jiang, P. Wu, Z. Liang, P. Q. Chen, X. Yuan, Y. Jia, J. Tu, C. Li, P. H. F. Ng, Q. Li. **KDD 2025 Benchmark & Dataset Track**. CCF-A.
- *Mesosopic Insights: Orchestrating Multi-Scale & Hybrid Architecture for Image Manipulation Localization*
X. Zhu, X. Ma, L. Su, **Z. Jiang**, B. Du, X. Wang, Z. Lei, W. Feng, C.-M. Pun, J. Zhou. **AAAI 2025**. CCF-A.

- **IMDL-BenCo: A Comprehensive Benchmark and Codebase for Image Manipulation Detection & Localization**
X. Ma[†], X. Zhu[†], L. Su[†], B. Du[†], **Z. Jiang[†]**, B. Tong[†], Z. Lei[†], X. Yang[†], C.-M. Pun, J. Lv, J. Zhou. **NeurIPS 2024 Benchmark Track – Spotlight**. [†] Co-first authors. CCF-A.
- **Beyond Visual Appearances: Privacy-Sensitive Objects Identification via Hybrid Graph Reasoning**
Z. Jiang, B. Tong, X. Du, A. Alhammedi, J. Zhou. arXiv:2406.12736, 2024.
- **TPTGAN: Two-Path Transformer-Based Generative Adversarial Network Using Joint Magnitude Masking and Complex Spectral Mapping for Speech Enhancement**
Z. Liu, **Z. Jiang**, W. Luo, Z. Fan, H. Di, Y. Long, H. Wang. **ICONIP 2023**, Lecture Notes in Computer Science. CCF-C.
- **IML-ViT: Benchmarking Image Manipulation Localization by Vision Transformer**
X. Ma, B. Du, **Z. Jiang**, A. Y. Al Hammadi, J. Zhou. arXiv:2307.14863, 2023.
- **Perceptual MAE for Image Manipulation Localization: A High-level Vision Learner Focusing on Low-level Features**
X. Ma, **Z. Jiang**, X. Xu, C.-M. Pun, J. Zhou. arXiv:2310.06525, 2023.

Research & Industry Experience

Kuaishou Technology

Beijing, China

RESEARCH INTERN, E-COMMERCE RECOMMENDATION CENTER

Oct. 2025 – Present

- Conduct research on LLM-enhanced recommender systems for short-video e-commerce and cross-domain content-to-commerce recommendation.
- Developed AIR, an LLM-driven framework that migrates semantic reasoning offline and composes user intent representations efficiently online for industrial-scale ranking systems.
- Work on user intent modeling, user persona construction, and semantic feature engineering for conversion-oriented recommendation.

Machinelab / DICALab, Sichuan University

Chengdu, China

RESEARCH ASSISTANT

Sep. 2022 – Jun. 2024

- Developed graph-based frameworks for privacy-sensitive object recognition and image manipulation localization.
- Participated in research projects funded by the National Natural Science Foundation of China and the National Key R&D Program of China.
- Co-authored multiple papers in top-tier conferences and benchmark tracks, including NeurIPS and AAAI.

School of Computing, National University of Singapore

Singapore

SUMMER SCHOOL PARTICIPANT

Aug. 2023

- Completed an intensive research program and a face-recognition project based on CNN feature extraction and similarity matching.

Selected Projects

AIR: Atomic Intent Reasoning for Industrial Cross-Domain Recommendation

First Author

KDD 2026 ADS TRACK

2025–2026

- Designed an LLM-driven intent reasoning framework for bridging short-video content behavior and e-commerce recommendation.
- Moved expensive LLM semantic inference to the offline stage and used online retrieval/composition to construct target-aware user intent representations.
- Achieved approximately 400× inference acceleration and improved Kuaishou E-commerce online GMV by +3.446% in large-scale A/B testing.

SUPERGLASSES / SUPERLENS: Smart-Glasses VQA Benchmark and Agent

First Author

CVPR 2026 FINDINGS

2026

- Built a real-world smart-glasses VQA benchmark with 2,422 egocentric image-question pairs across 14 domains and 8 query types.
- Proposed SUPERLENS, a multimodal RAG agent integrating object detection, query decomposition, and multimodal web search.

HiBench: Benchmark for Hierarchical Reasoning

First Author & Team Leader

KDD 2025 BENCHMARK & DATASET TRACK

2025

- Designed and led the first comprehensive benchmark for hierarchical structure reasoning in LLMs, covering 30 tasks and 39,519 queries across six scenarios.
- Led evaluation framework design, key implementation, dataset/toolkit release, and multi-institutional collaboration.

Meta CRAG-MM: Multimodal Retrieval Challenge

Team Leader

KDD Cup 2025

2025

- Led a team to achieve **12th place globally** among hundreds of international teams.
- Designed multimodal RAG pipelines with domain routing, search routing, and multi-hop reasoning for knowledge-intensive VQA.

IMDL-BenCo: Benchmark for Image Manipulation Detection & Localization

Co-first Author

NEURIPS 2024 BENCHMARK TRACK

2024

- Developed a comprehensive benchmark and modular codebase for image manipulation detection and localization.
- Implemented GPU-accelerated evaluation metrics and contributed to benchmark design and manuscript writing.

Invited Talks

Reasoning Day @ KDD 2025

Toronto, ON, Canada

UNDERSTANDING HIERARCHICAL DATA WITH LARGE LANGUAGE MODELS: RAG, STRUCTURAL REASONING, AND FUTURE

Aug. 2025

DIRECTIONS

- Invited talk on hierarchical data understanding, RAG systems, structural reasoning, and future directions in reasoning-enhanced AI systems.

Honors & Awards

- 2025 **12th Place (Global)**, KDD Cup 2025 – Meta CRAG-MM Multimodal Retrieval Challenge
- 2024 **Spotlight Award**, NeurIPS 2024 Benchmark Track – IMDL-BenCo (Co-first Author)
- 2024 **Outstanding Graduate**, Sichuan University & Sichuan Province
- 2023 **Provincial First Prize**, China Collegiate Computer Design Competition
- 2023 **Tencent Scholarship**, Top 2% Awardee, Sichuan University
- 2023 **A-Level Certificate**, Comprehensive Quality Evaluation
- 2022 **Top 1% Scholarship**, Comprehensive First-Class Scholarship, Sichuan University
- 2022 **Outstanding Student Award**, Top 5%, Sichuan University

Toronto, Canada
Vancouver, Canada
Sichuan, China
Sichuan, China
Sichuan, China
Sichuan, China
Sichuan, China
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Professional Service & Teaching

Conference & Journal Reviewing

AoE

REVIEWER

2023 – Present

- Reviewed for TIP, CVPR, ICCV, ECCV, NeurIPS, KDD, AAAI, and IoTJ.

Frontiers in Artificial Intelligence; Frontiers in Big Data

AoE

TOPIC COORDINATOR

2025 – 2026

- Topic Coordinator for Frontiers in Artificial Intelligence and Frontiers in Big Data.

The Hong Kong Polytechnic University

Hong Kong SAR

TEACHING ASSISTANT

2024 – 2025

- Assisted courses including Artificial Intelligence (COMP4431), NLP Practicum (COMP5423), and Database System (COMP2411).

Skills

Research Areas	Large Language Models, Retrieval-Augmented Generation, Recommender Systems, Multimodal Agents, Cross-Domain Recommendation, Computer Vision
Frameworks & Tools	PyTorch, Hugging Face, NumPy, OpenCV, Docker, Git, Anaconda, LaTeX
Languages	Mandarin Chinese (native), English (fluent, IELTS 6.5)